

LAB 9 Gym Management

#DDL

```
create database gym_management;
```

```
create table members(  
    id int primary key ,  
    full_name varchar(50) not null ,  
    email varchar(50) not null unique ,  
    phone varchar(10) unique ,  
    gender char(1) not null check ( gender='f' or gender='m'),  
    date_of_birth date not null ,  
    created_at datetime default current_timestamp  
);
```

```
create table membership_plans (  
    id int primary key ,  
    name varchar(20) not null unique ,  
    price double not null check ( price>0 ),  
    duration_months int not null check (duration_months > 0),  
    description varchar(100),  
    status varchar(11)not null check (status = 'available' or status = 'unavailable'),  
    created_at datetime default current_timestamp  
);
```

```
create table trainers (  
    id int primary key ,  
    full_name varchar(50) not null ,  
    email varchar(50) not null unique ,  
    phone varchar(10) unique ,  
    specialty varchar(30) not null ,  
    experience_years int check (experience_years >= 0),
```

```
created_at datetime default current_timestamp
);
```

```
create table member_subscriptions (
id int primary key ,
member_id int not null ,
foreign key (member_id) references members(id),
membership_plan_id int not null ,
foreign key (membership_plan_id) references membership_plans(id),
start_date date not null ,
end_date date not null check (end_date > start_date),
status varchar(7) not null check (status = 'active' OR status = 'expired' or status =
'pending'),
created_at datetime default current_timestamp
);
```

```
create table classes (
id int primary key,
name varchar(30) not null,
trainer_id int not null,
foreign key (trainer_id) references trainers(id),
class_date date not null,
start_time time not null,
end_time time not null check (end_time > start_time),
capacity int not null check (capacity > 0),
created_at datetime default current_timestamp
);
```

```
create table bookings (
id int primary key,
member_id int not null,
foreign key (member_id) references members(id),
class_id int not null,
```

```
foreign key (class_id) references classes(id),  
booking_date datetime default current_timestamp,  
status varchar(9) not null check (status = 'booked' or status = 'cancelled')  
);
```

```
create table reviews (  
id int primary key,  
member_id int not null,  
foreign key (member_id) references members(id),  
trainer_id int not null,  
foreign key (trainer_id) references trainers(id),  
rating int not null check (rating between 1 and 5),  
comment varchar(200),  
created_at datetime default current_timestamp  
);
```

```
create table payments (  
id int primary key,  
subscription_id int not null,  
foreign key (subscription_id) references member_subscriptions(id),  
amount double not null check (amount > 0),  
payment_method varchar(10) not null check (payment_method = 'cash' or  
payment_method = 'card'),  
payment_status varchar(7) not null check (payment_status = 'paid' or  
payment_status = 'failed'),  
payment_date datetime default current_timestamp  
);
```

#DML

#table member

```
insert into members values(1, 'Lama Alharbi', 'lama@gmail.com', '0501111111', 'f',  
'1997-08-09', default);
```

```
insert into members values(2, 'Sara Ahmed', 'sara@gmail.com', '0502222222', 'f',  
'1998-05-15', default);
```

```
insert into members values(3, 'Khalid Ali', 'khalid@gmail.com', '0503333333', 'm',  
'1995-03-10', default);
```

```
insert into members values(4, 'Nora Fahad', 'nora@gmail.com', '0504444444', 'f',  
'2000-11-25', default);
```

```
insert into members values(5, 'Omar Mohammed', 'omar@gmail.com',  
'0505555555', 'm', '1996-07-08', default);
```

```
#table membership_plans
```

```
insert into membership_plans values(1, 'Basic', 150, 1, 'Basic monthly plan',  
'available', default);
```

```
insert into membership_plans values(2, 'Standard', 400, 3, 'Three months plan',  
'available', default);
```

```
insert into membership_plans values(3, 'Premium', 700, 6, 'Six months plan',  
'available', default);
```

```
insert into membership_plans values(4, 'Annual', 1200, 12, 'One year membership',  
'available', default);
```

```
insert into membership_plans values(5, 'Student', 100, 1, 'Special plan for students',  
'unavailable', default);
```

```
#table trainers
```

```
insert into trainers values(1, 'Fahad Salem', 'fahad@gym.com', '0511111111',  
'Bodybuilding', 6, default);
```

```
insert into trainers values(2, 'Reem Ali', 'reem@gym.com', '0522222222', 'Yoga', 4, default);
```

```
insert into trainers values(3, 'Ahmed Omar', 'ahmed@gym.com', '0533333333', 'CrossFit', 7, default);
```

```
insert into trainers values(4, 'Huda Sami', 'huda@gym.com', '0544444444', 'Pilates', 3, default);
```

```
insert into trainers values(5, 'Saud Nasser', 'saud@gym.com', '0556666666', 'Boxing', 8, default);
```

```
# table member_subscriptions
```

```
insert into member_subscriptions values(1, 1, 1, '2026-09-01', '2026-10-01', 'active', default);
```

```
insert into member_subscriptions values(2, 2, 3, '2026-09-10', '2027-03-10', 'active', default);
```

```
insert into member_subscriptions values(3, 3, 2, '2026-09-15', '2026-12-15', 'pending', default);
```

```
insert into member_subscriptions values(4, 4, 4, '2025-08-01', '2026-08-01', 'expired', default);
```

```
insert into member_subscriptions values(5, 5, 1, '2026-09-18', '2026-10-18', 'pending', default);
```

```
# classes
```

```
insert into classes values(1, 'Strength Training', 1, '2026-09-20', '17:00:00', '18:00:00', 20, default);
```

```
insert into classes values(2, 'Morning Yoga', 2, '2026-09-21', '08:00:00', '09:00:00',  
15, default);
```

```
insert into classes values(3, 'CrossFit Challenge', 3, '2026-09-22', '18:00:00',  
'19:30:00', 18, default);
```

```
insert into classes values(4, 'Pilates Class', 4, '2026-09-23', '16:00:00', '17:00:00',  
12, default);
```

```
insert into classes values(5, 'Boxing Training', 5, '2026-09-24', '20:00:00',  
'21:30:00', 16, default);
```

```
# bookings
```

```
insert into bookings values(1, 1, 2, default, 'booked');
```

```
insert into bookings values(2, 2, 1, default, 'booked');
```

```
insert into bookings values(3, 3, 3, default, 'cancelled');
```

```
insert into bookings values(4, 4, 4, default, 'booked');
```

```
insert into bookings values(5, 5, 5, default, 'cancelled');
```

```
# reviews
```

```
insert into reviews values(1, 1, 2, 5, 'Great trainer', default);
```

```
insert into reviews values(2, 2, 1, 4, 'Very good training', default);
```

```
insert into reviews values(3, 3, 3, 5, 'Excellent experience', default);
```

```
insert into reviews values(4, 4, 4, 4, 'Good class and trainer', default);
```

insert into reviews values(5, 5, 5, 3, 'Good but can be better', default);

payments

insert into payments values(1, 1, 150, 'card', 'paid', default);

insert into payments values(2, 2, 700, 'card', 'paid', default);

insert into payments values(3, 3, 400, 'cash', 'failed', default);

insert into payments values(4, 4, 1200, 'card', 'paid', default);

insert into payments values(5, 5, 150, 'card', 'failed', default);

UPDATE

update members set phone = '0509999999' where id = 1;

update membership_plans set price = 650 where id = 3;

update trainers set experience_years = 8 where id = 3;

update member_subscriptions set status = 'active' where id = 3;

update classes set capacity = 25 where id = 1;

update bookings set status = 'cancelled' where id = 2;

update reviews set rating = 5 where id = 2;

update payments set payment_status = 'paid' where id = 3;

DELETE

delete from bookings where id = 5;

delete from reviews where id = 5;

delete from payments where id = 5;

delete from payments where subscription_id = 3;

delete from bookings where member_id = 3;

delete from reviews where member_id = 3;

delete from member_subscriptions where member_id = 3;

delete from members where id = 3;

#DQL

select * from members;

select * from membership_plans;

select * from trainers;

select * from member_subscriptions;

select * from classes;

select * from bookings;

select * from reviews;


```
select * from payments;
```

```
# business queries
```

```
select * from membership_plans where status = 'available';
```

```
select * from member_subscriptions where status = 'active';
```

```
select * from member_subscriptions where status = 'pending';
```

```
select * from bookings where status = 'booked';
```

```
select * from classes where capacity > 15;
```

```
select * from trainers where experience_years >= 5;
```

```
select * from reviews where rating >= 4;
```

```
select * from payments where payment_status = 'paid';
```

```
select * from payments where payment_status = 'failed';
```

```
select * from membership_plans order by price asc;
```

```
select * from trainers order by experience_years desc;
```